

IMAGE PROCESSING TECHNIQUES

Extended Class Activities Worksheet | UG Level | Activity 15: Corner & Interest Point Detection

WORKSHEET SOLUTIONS

1. 1. Define Interest Point.

An interest point (or keypoint) is a point in an image that possesses a clear, mathematically well-defined position and significant local feature variation in 2D space, making it stable, distinctive, and repeatable under transformation.

2. 2. Name two corner detection algorithms.

Harris Corner Detector and Shi-Tomasi Corner Detector (or FAST - Features from Accelerated Segment Test).

3. 3. Why are corners more stable than edges for tracking?

Corners experience significant intensity variations in all 2D directions, resolving the *aperture problem* inherent to edges (where motion parallel to an edge is ambiguous). This distinct 2D localization makes corners far more robust against rotation, translation, and view angle changes across frames.

4. 4. In which applications are interest points used?

Interest points are widely used in:

- Image Stitching and Panorama Generation
- Object Detection and Tracking in Video Sequences
- 3D Reconstruction and Structure from Motion (SfM)
- Visual Simultaneous Localization and Mapping (V-SLAM) for Autonomous Systems